



Developed by Y. Kon (Geological Survey of Japan, AIST)
Version 1.5.5 (2026/8/26)

Quick Manual for Offline Targeting



Prepare a sample image and download PiXY

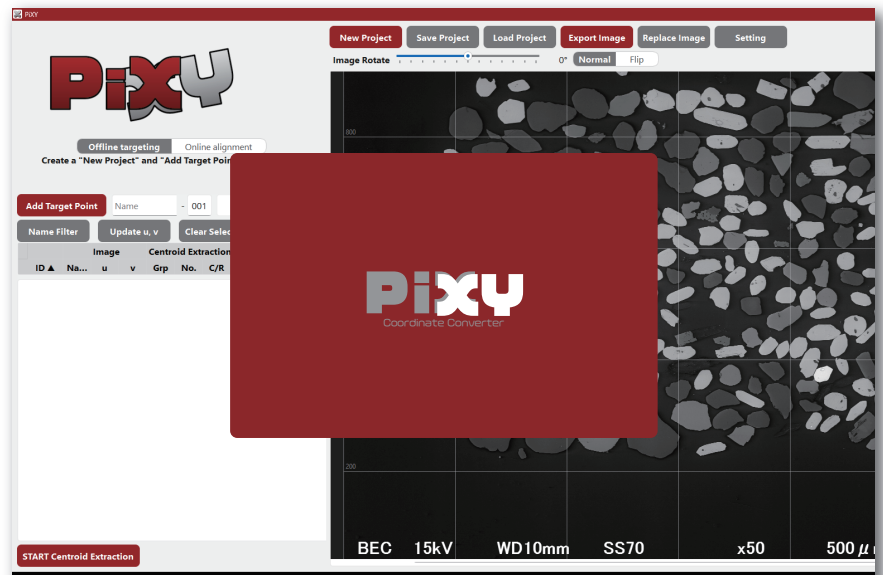
Photographs, scans, or microscope images can be used.

Download the latest PiXY from GitHub (<https://github.com/YoshiakiKON/PiXY/releases>).



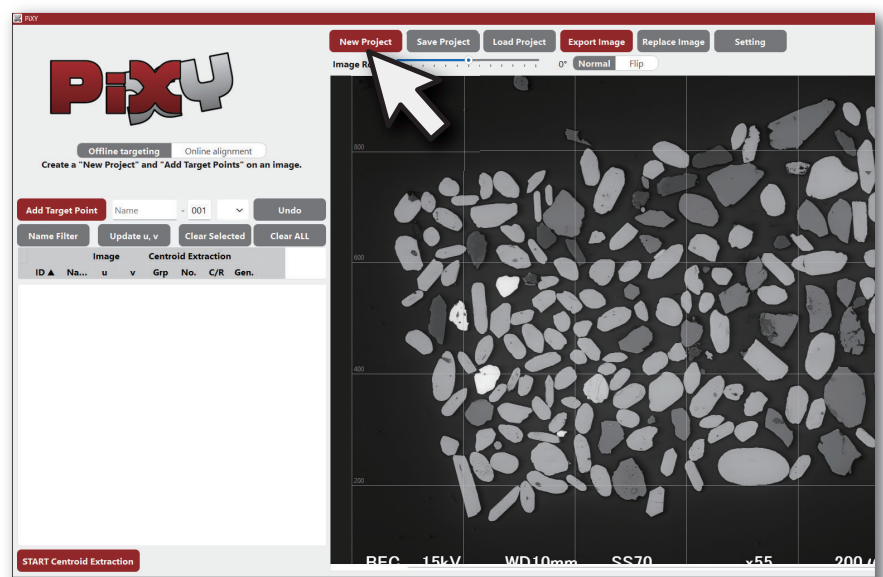
Launch PiXY.exe at office to Start "Offline Targeting"

If PiXY.exe is blocked by Windows security settings, try the distributed folder version, modify the Windows security settings, or launch PiXY from Python.



Start 'New Project' and load a sample image

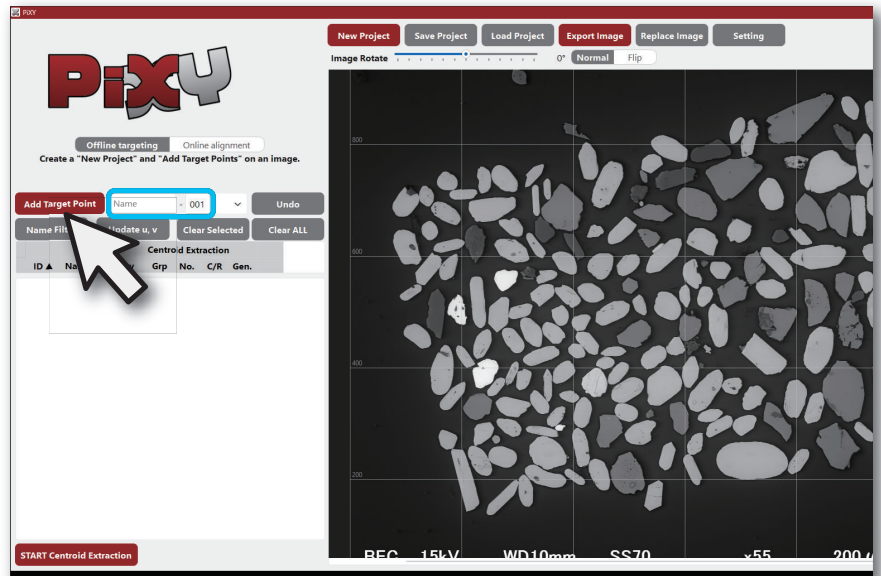
PiXY supports .png, .jpg (.jpeg),
.bmp, and .tif (.tiff) files.





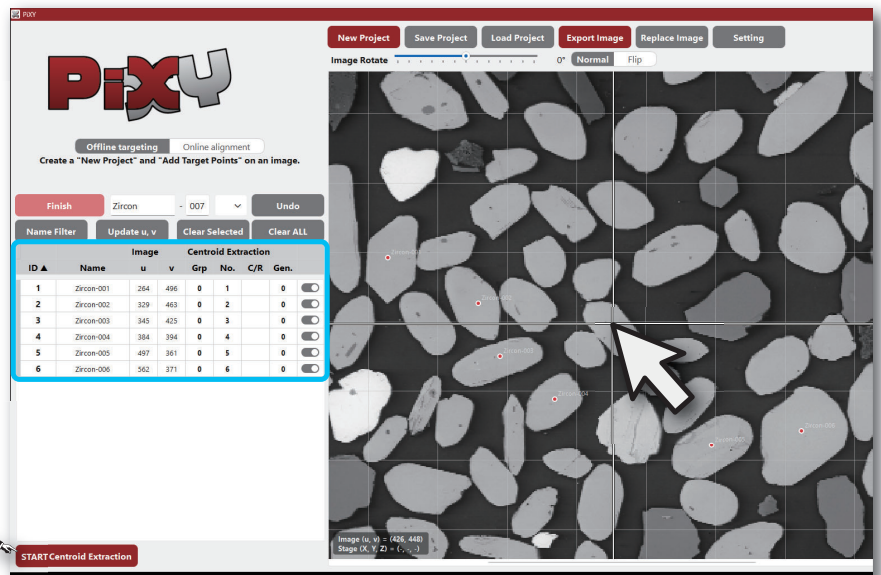
Click `Add Target Point` to start targeting

Each new Target Point is assigned the name entered in the "Name" field, followed by an automatically incremented number.



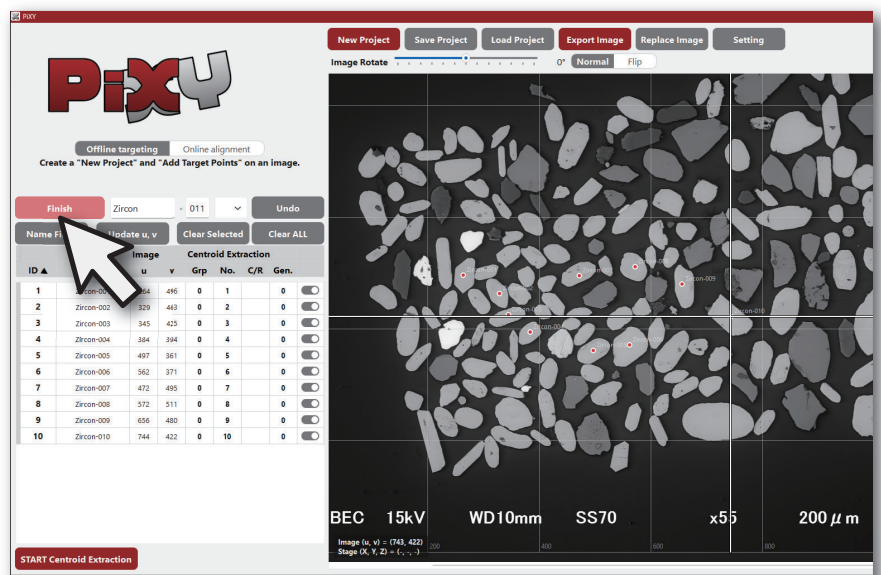
Click the sample image to add a Target Point to the Target List

For particle samples, you can also try automatic 'Centroid Extraction' mode.



Click `Finish` to complete Target Point addition

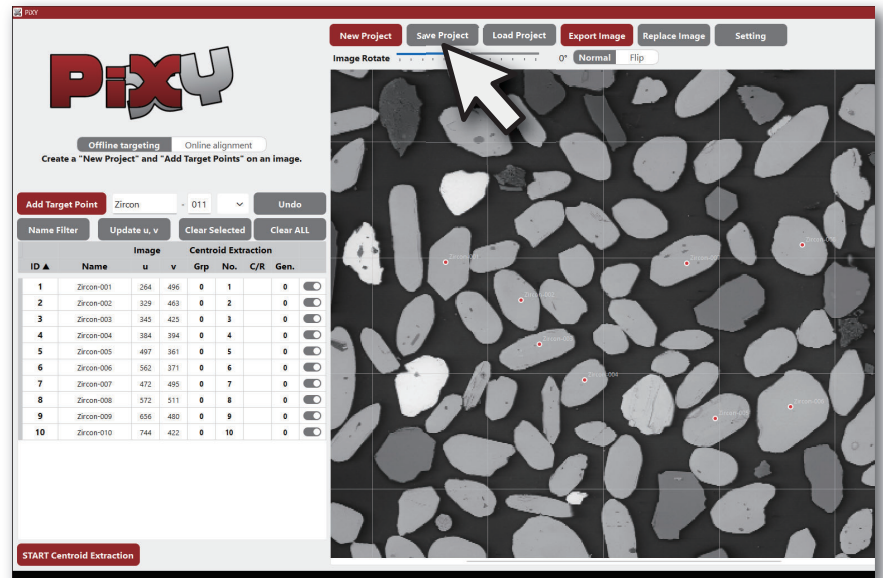
After adding at least one Target Point, the mode automatically ends when the cursor moves outside the sample image.



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'Save Project' to complete "Offline Targeting"

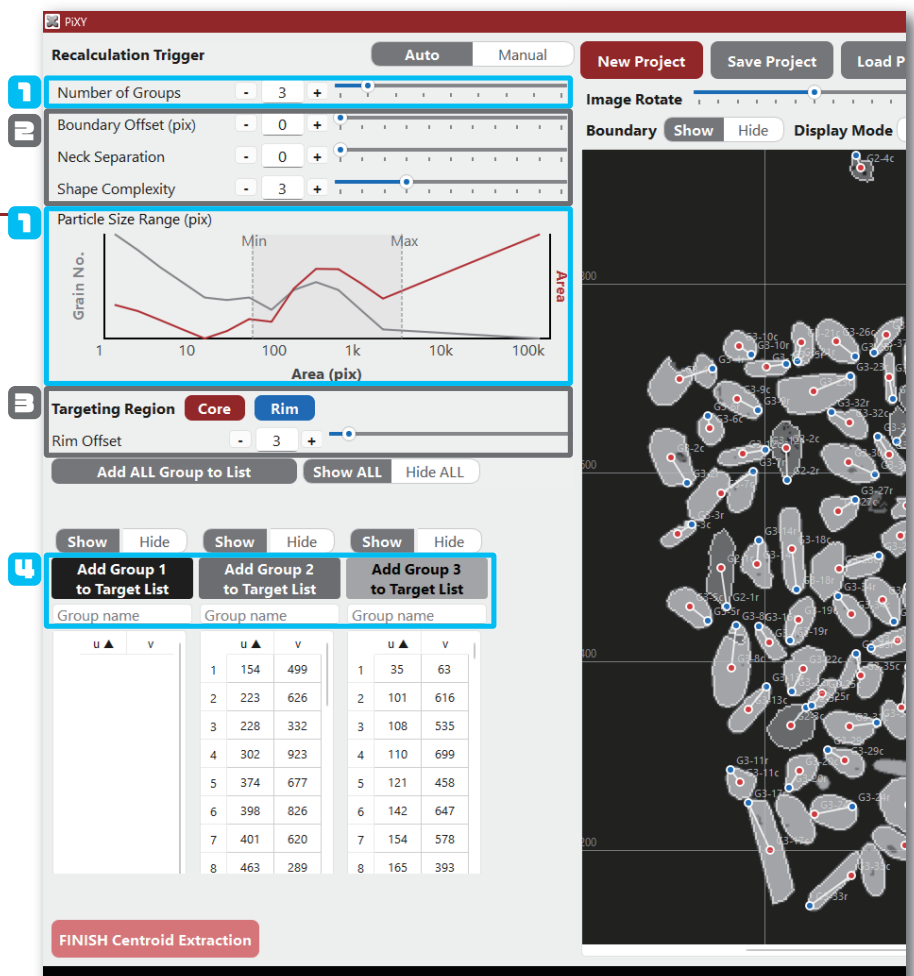
The loaded image and all entered Target Points are saved in a project file (.pixy).



Centroid Extraction Mode

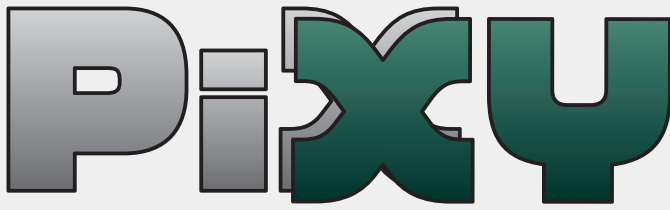
Extract Target Points from the image

- 1 Set the number of groups for K-means posterization and the particle size range to detect particles.
- 2 [advanced] Fine-tune the boundary-, neck-, and shape-related parameters to separate adjacent particles.
- 3 [advanced] Enable core-rim detection and set the target distance from the rim boundary.
- 4 Add the desired particle group to the Target List.



Citation

Kon, Y. (2026) "PiXY: Pixel to stage-XY Coordinate Converter". Zenodo. Available at: <https://doi.org/10.5281/zenodo.18174474>.



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Quick Manual for Online Alignment



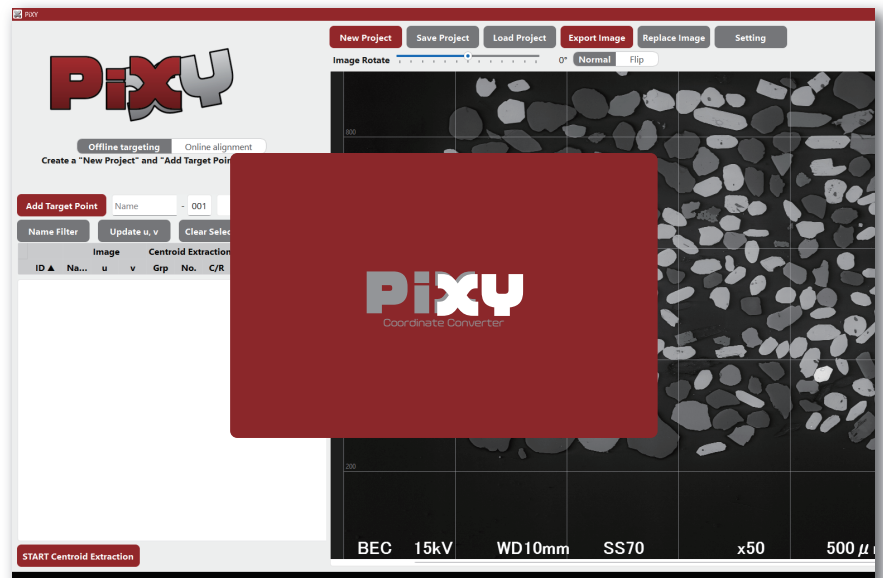
Complete Offline Targeting and prepare the instrument

Ensure that the project file (.pixy) are available on the PC used for Online Targeting.



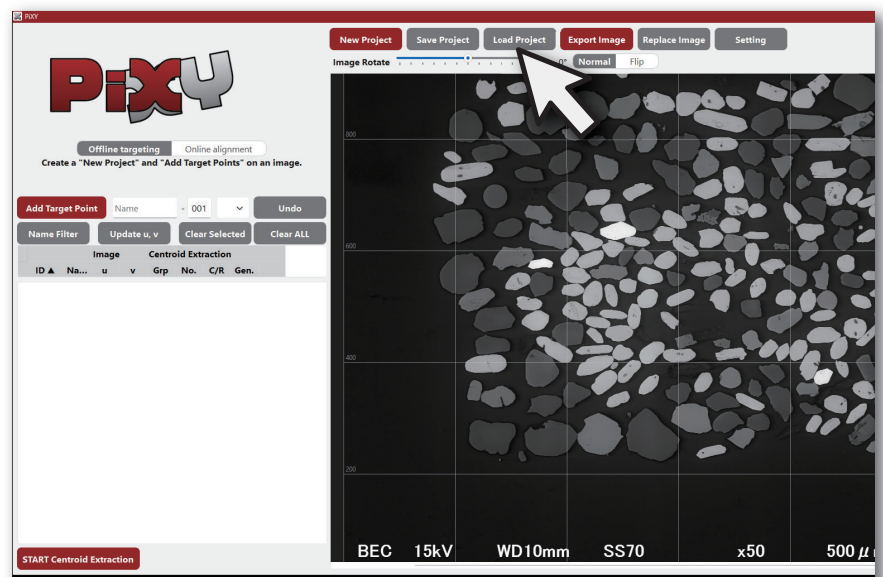
Launch PiXY.exe at laboratory to Start "Online Alignment"

Coordinate transfer is easier when PiXY is run on the instrument control PC. When using your own PC in the laboratory, the exported coordinate file must be copied to the instrument control PC.



Click 'Load Project' and select a project file (.pixy)

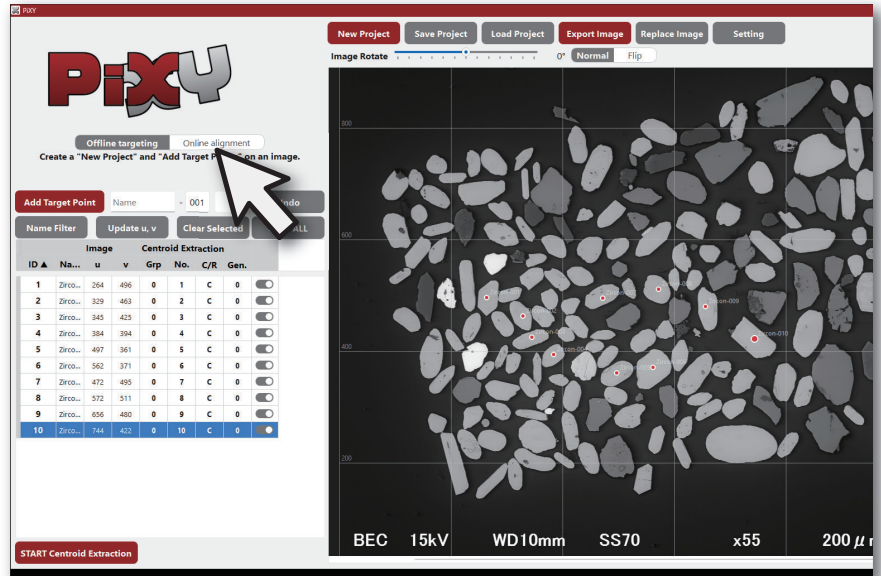
If PiXY is already running in Offline Targeting mode and the same PC is brought into the laboratory, skip these steps and proceed to Step 3.





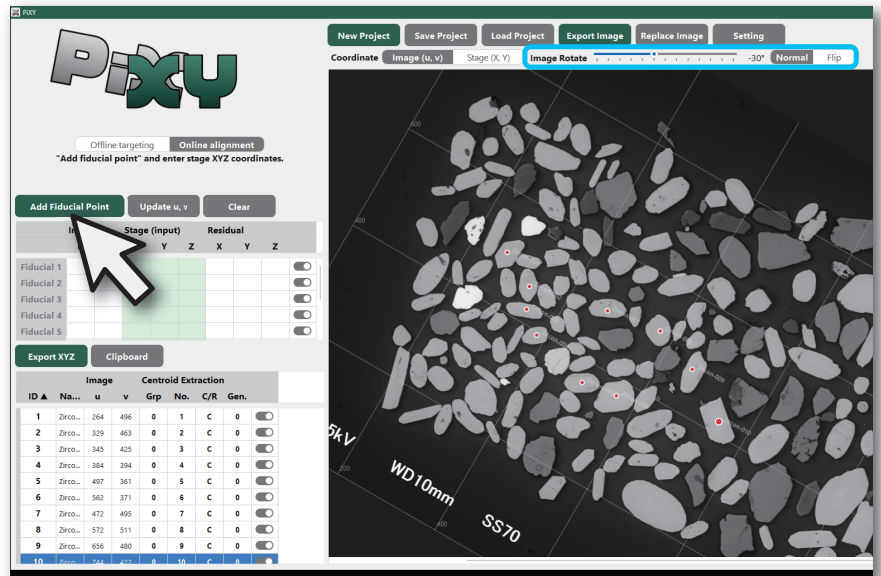
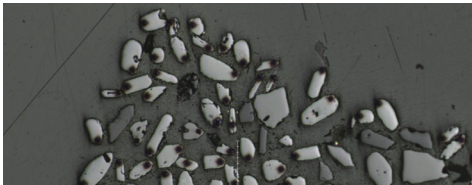
Click "Online Alignment"

The UI then switches to Online Targeting mode.



Click 'Add Fiducial Point'

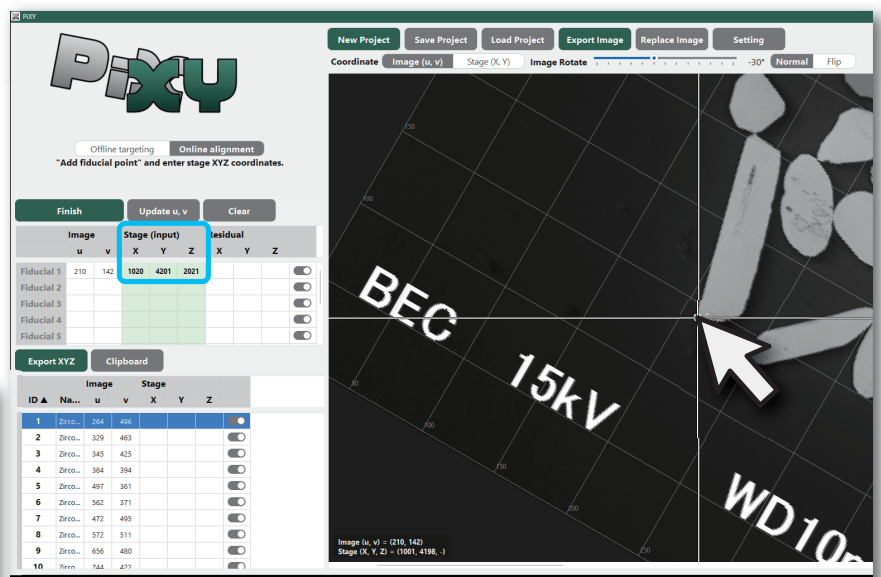
Match the image to the sample orientation to find fiducials more easily.



Select a Fiducial Point on the image and enter its XYZ coordinates on the instrument

Locate the
corresponding point on the
instrument and
record its stage
coordinates.

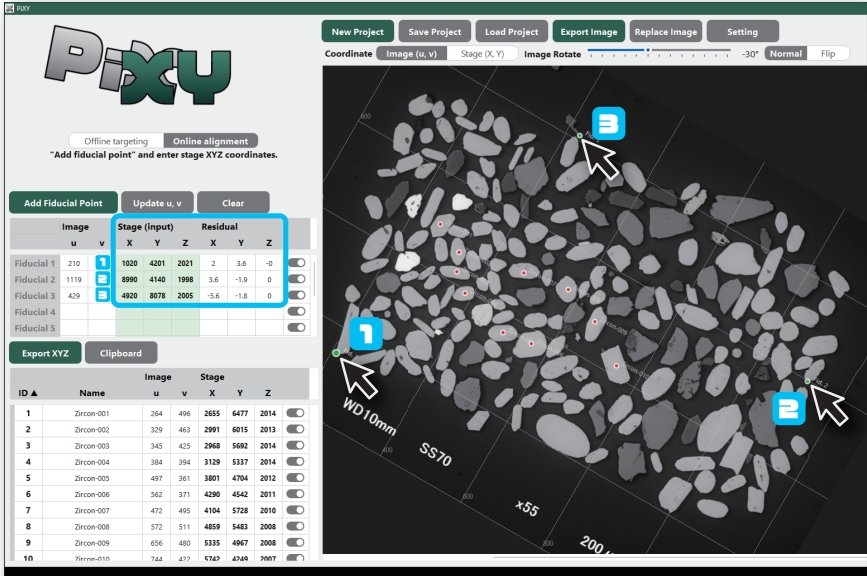
X: 1020
Y: 4201
Z: 2021



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Continue adding Fiducial Points

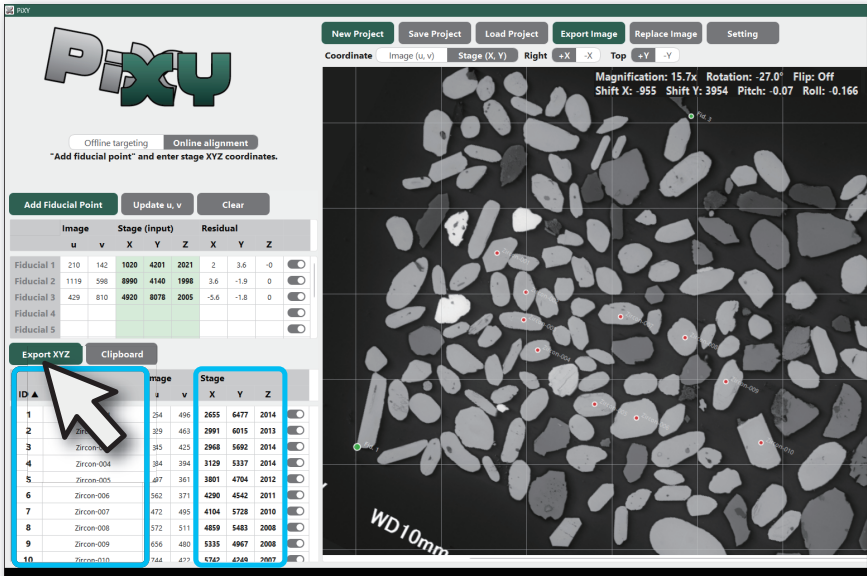
Each new Fiducial Point refines the transformation. Continue adding points while checking the residuals (minimum 3 points).



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Export the calculated XYZ stage coordinates

Export the XYZ coordinates in the Target List as a .txt or .csv file, or copy them to the clipboard.



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Target Coordinates Ready to Use

Export the Target List coordinates as a CSV file or copy them to the clipboard for use in instrument software.

	A	B	C	D	E	F
1	No	Name	X	Y	Z	
2		1 Zircon-001	2655.41	6476.82	2013.501	
3		2 Zircon-002	2991.485	6014.662	2013.089	
4		3 Zircon-003	2967.67	5692.334	2013.55	
5		4 Zircon-004	3129.376	5336.854	2013.513	
6		5 Zircon-005	3800.531	4703.528	2012.34	
7		6 Zircon-006	4289.945	4541.545	2011.119	
8		7 Zircon-007	4103.856	5728.109	2010.217	
9		8 Zircon-008	4859.005	5482.201	2008.228	

Citation

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